

3.2.10 Structured programming and subroutines (procedures and functions)

Content	Additional information
Understand the concept of subroutines.	Students should know that a subroutine is a named 'out of line' block of code that may be executed (called) by simply writing its name in a program statement.
Explain the advantages of using subroutines in programs.	
Describe the use of parameters to pass data within programs.	Students should be able to use subroutines that require more than one parameter. Students should be able to describe how data is passed to a subroutine using parameters.
Use subroutines that return values to the calling routine.	Students should be able to describe how data is passed out of a subroutine using return values.
Know that subroutines may declare their own variables, called local variables, and that local variables usually: • only exist while the subroutine is executing • are only accessible within the subroutine.	
Use local variables and explain why it is good practice to do so.	
Describe the structured approach to programming.	Students should be able to describe the structured approach including modularised programming, clear well-documented interfaces (local variables, parameters) and return values. Teachers should be aware that the terms arguments and parameters are sometimes used but in examinable material we will use the term parameter to refer to both of these.
Explain the advantages of the structured approach.	